



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Charge and Current — $Q = It$

One equation, one trap, and one number that puts the whole thing in perspective.

Key concept

CURRENT IS A FLOW OF CHARGE

Current is not “electricity”. It is the **rate at which charge flows** — how much charge goes past a point each second.

$$\text{charge flow} = \text{current} \times \text{time} \quad Q = It$$

Q = charge flow, in **coulombs (C)** · I = current, in **amps (A)** · t = time, in **seconds (s)**

Rearranged: $I = \frac{Q}{t}$ and $t = \frac{Q}{I}$

So how much is one coulomb?

The charges doing the flowing in a metal wire are **electrons**, and a single electron carries a fantastically small charge:

$$e = 1.6 \times 10^{-19} \text{ C}$$

So a charge of Q coulombs is made up of

$$\text{number of electrons} = \frac{Q}{e}$$

This is worth doing once, because otherwise a coulomb is just a word. The numbers on this sheet — 34 C, 108 C — sound modest. They are not. *You will not be asked for the value of e in a Combined Science exam.*

And how fast do the electrons move?

Slowly. In a copper wire carrying a normal current the electrons **drift** along at well under a millimetre per second — slower than a snail.

Which leaves a puzzle, and question 9 asks you to solve it: **if the electrons crawl, why does a lamp light the *instant* you close the switch?**

The trap, and it is the one that costs marks. t must be in **seconds**. Real questions give you *minutes*, or hours. Convert **first**, and write the converted number down so a marker can see it.

$$1 \text{ minute} = 60 \text{ s} \quad \cdot \quad 2 \text{ minutes} = 120 \text{ s} \quad \cdot \quad 1 \text{ hour} = 3600 \text{ s}$$

Worked example

A current of 0.25 A flows for **3 minutes**. Calculate the charge that flows, and the number of electrons that represents.

Convert first. $t = 3 \times 60 = 180 \text{ s}$ $Q = It = 0.25 \times 180 = \mathbf{45 \text{ C}}$

Then the electrons. $\frac{Q}{e} = \frac{45}{1.6 \times 10^{-19}} = \mathbf{2.8 \times 10^{20} \text{ electrons}}$

Using $t = 3$ would give 0.75 C — sixty times too small, and no method mark for a conversion you never wrote

down. And 2.8×10^{20} is 280 million million million electrons, for a quarter of an amp over three minutes.

1. State what is meant by *electric current*. [1]

2. A doorbell draws a current of 0.85 A for 40 seconds. Calculate the charge flow. **Give your answer to 2 significant figures.** [3]

3. A charge of 624 C flows through a motor in 130 s. Calculate the current. [2]

4. A phone charger supplies a current of 0.90 A for **2 minutes**. Calculate the charge that flows. [3]

5. A torch bulb carries a current of 0.24 A. How long must it be switched on for 72 C of charge to flow? Give your answer **in minutes**. [3]

6. A torch bulb carries a current of 0.30 A for 60 seconds. Calculate the charge that flows, and then the number of electrons. [3]

7. A student says: “A bigger current always means more charge has flowed.” Explain why this is not necessarily true. [2]

8. A single closed loop carries a current of 2.0 A. Explain why the same charge flows past every point in that loop in one minute. [2]

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9. Electrons in a copper wire drift along at less than a millimetre per second. Explain why a lamp lights immediately when you close the switch. [3]
